

SURGE END-TERM REPORT

Affective computing on Indian Demographics.

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Introduction:

The aim of the project is to develop multimodal computational models for the affect (emotion) prediction in the Indian setting. Most of the existing emotion recognition datasets are situated in the Western setting i.e., videos depicting western culture and languages. However, humans express emotions differently in different cultures. We plan to address this by creating a new multimodal (video with audio and text transcripts) affective corpus to capture the Indian culture and demographics and subsequently, develop deep learning techniques for recognizing affect in the Indian setting (e.g., in Indian videos).

We further plan to study the effect of culture, language, and other demographics on emotion prediction systems. An analysis of change in emotions in different demographics would be helpful to understand the effect of cultural cues present in the corpus.

Work done: We have nearly completed the first stage of the project, continuing from the UGP, and will start the modeling process soon. The work done so far is as follows:

2)Creation of open source user-control enabled label-studio Development of a web interface for annotating India-specific TV series / Movies:

- We have modified [Label Studio](#)(an open-source data labeling tool), to suit our needs for the project. We chose Label studio because it has a customizable UI and a simple interface.
- The free version of Label Studio does not have any user-based hierarchy. Any annotator can delete other annotations, or even delete the whole project. So we made changes to the source code by removing the danger buttons like delete and Settings button.
- We have released this modified version on Github ([here](#)), which will be useful for many others also working on Machine Learning projects which require a ready-made flexible annotation tool like label-studio with user control.
- We also wrote a [blog](#) with a full tutorial video explaining how to set it up.
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- We created XML code to have the labeling interface display the context and utterance video. The tool allows annotators to choose the tags “Humour”, and “Sarcasm”, and select an emotion as shown in the image.
- The XML code was modified so that every emotion has a rating (from 1-8) in the form of stars, that can be selected.
- We fixed the issues that arose due to different aspect ratios across computers

☐
☒ video

#6
AD Admin #D-vtJ
1/1

☐
☒ 0

/data/local-files/?
d=data/tejasr20/dat
_ep1/kota_factory_1

☐
☒ 0

/data/local-files/?
d=data/tejasr20/dat
_ep1/kota_factory_1

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/data/local-files/?
d=data/tejasr20/dat
_ep1/kota_factory_2

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☒ 0

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d=data/tejasr20/dat
_ep1/kota_factory_2

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/data/local-files/?
d=data/tejasr20/dat
_ep1/kota_factory_2

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☒ 0

/data/local-files/?
d=data/tejasr20/dat
_ep1/kota_factory_2

What type of Emotion?

☒ Positive^[4]
☐ Negative^[5]
☐ Neutral^[6]

Fine Grained Positive Emotion?

☒ Admiration^[7]

★ ★ ★ ★ ★ ★ ★ ★

☐ Approval^[9]

★ ★ ★ ★ ★ ★ ★ ★

☐ Gratitude^[4]

★ ★ ★ ★ ★ ★ ★ ★

☐ Amusement^[6]

★ ★ ★ ★ ★ ★ ★ ★

☐ Love^[8]

★ ★ ★ ★ ★ ★ ★ ★

☐ Optimism^[4]

★ ★ ★ ★ ★ ★ ★ ★

☐ Joy^[9]

★ ★ ★ ★ ★ ★ ★ ★

☒ Caring^[3]

★ ★ ★ ★ ★ ★ ★ ★

☐ Excitement^[4]

★ ★ ★ ★ ★ ★ ★ ★

Task #6

We have set up the annotation tool on the image2 server, and plan to get it annotated so that the modeling portion can start.

1) Creation of Indian multimodal emotion recognition dataset:

- We conducted web searches for various Indian movies and TV series with emotion-specific information from YouTube.
- For dividing the videos into utterances, we used the transcript provided by YouTube that was present for very few movies/TV series. We used two python scripts, one of which extracts the start and end times from the transcript file, and the other breaks the videos into segments according to those times.
- Our dataset currently has videos from Kota Factory, PK, Nil Battey Sannata, and Ahalya. The languages we have so far are Hindi, English, Bhojpuri, and Bengali. The dataset is still under construction. We have faced difficulties in getting videos on Youtube that have transcripts and screenplays available for other languages.
- We cleaned up the dataset manually, by removing songs, narrations, and others. After cleanup, we were left with a total of 2439 utterance videos.
- Next, we edited the transcripts scene by scene and used them to create the context videos for each utterance. (The context video is needed to give the annotator some context with respect to the scene).
- We aligned each utterance with the corresponding context video in JSON files that will be imported onto the annotation tool to be annotated.